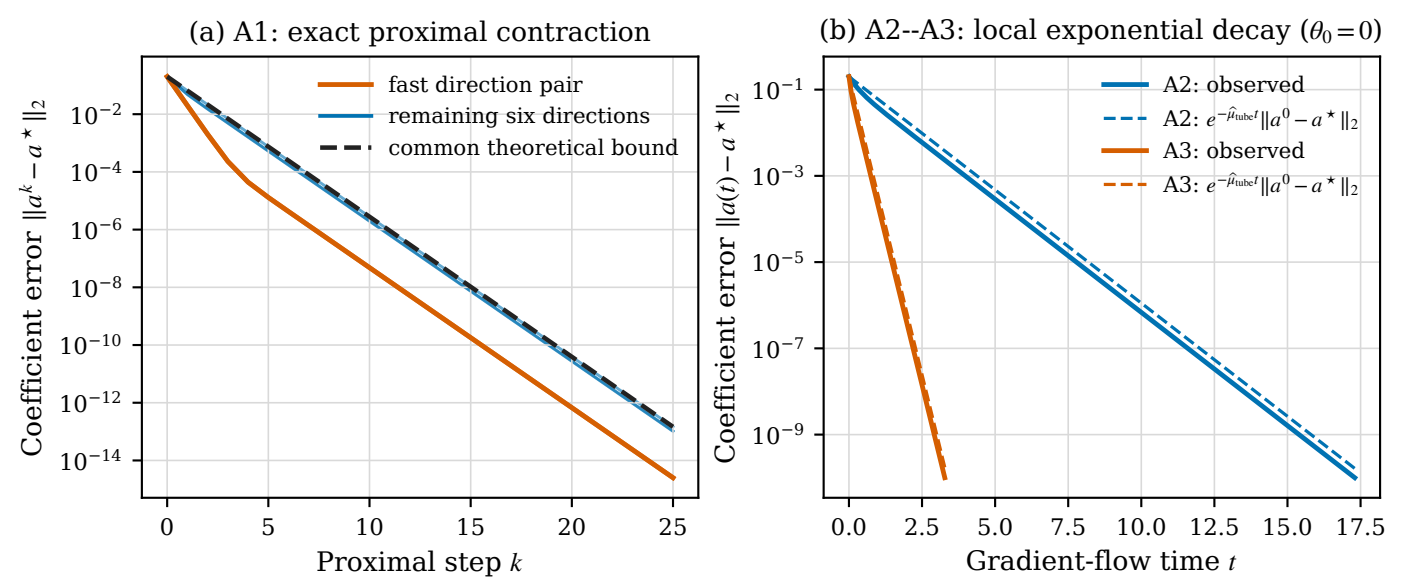
(a) A1: exact proximal contraction
fast direction pair
remaining six directions
common theoretical bound
Coefficient error $\|a^k - a^\star\|_2$
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
0
5
10
15
20
25
Proximal step k
(b) A2--A3: local exponential decay ($\theta_0 = 0$)
A2: observed
A2: $e^{-\widehat{\mu}_{\mathrm{tube}} t}\|a^0 - a^\star\|_2$
A3: observed
A3: $e^{-\widehat{\mu}_{\mathrm{tube}} t}\|a^0 - a^\star\|_2$
Coefficient error $\|a(t) - a^\star\|_2$
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Gradient-flow time t